How to react in liquid markets: The effect of precautionary interventions on interbank markets in emerging countries

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Abstract:

We examine the shock of the global crisis on interbank markets in emerging countries and attempt to assess the impact of macroeconomic and financial sector policy announcements on spread changes in six Central and Eastern European (CEE) countries. We find that interbank markets in emerging countries have reacted to the global events of 2007-2011; however, the policy initiatives have not been successful in stabilizing the situation. In particular, we show that the actions of central banks have not been effective when liquidity is not a concern. Instead, financial sector policy instruments seem to work better. The weak evidence on international interventions demonstrates their important impact on the stability of interbank markets. Regulators in emerging countries should introduce new instruments for interbank markets where liquidity is not a concern.

Keywords: financial crisis, interbank market, interventions, policy announcements, event study

JEL: G01, G14, G15, G18, E65, E63

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1. Introduction

During the global financial crisis of 2008-2011, the interbank markets in advanced countries stopped to function efficiently. Banks, unsure about the depth of the problems of other banks' balance sheets, were simply unwilling to lend to each other without significant accommodation for counterparty risk. As a result, the interbank spreads increased significantly during this time and the volume of transaction significantly decreased. The existing literature explains this "phenomena" mainly by liquidity and counterparty credit risk, which existed in the interbank market during this time (McAndrews et al., 2009; Schwarz, 2009; Poskitt, 2011; Ait-Sahalia, 2012). In addition, the studies also claim that banks became extremely risk-averse and preferred to accumulate liquidity rather than to transfer it on the market (Angelini et al., 2011; Acharya and Merrouche, 2012). To counteract, the lack of liquidity in the interbank market, central banks in most of the advanced countries stepped in and decided to provide liquidity to the system through a wide set of instruments.

The interbank markets in emerging countries have also experiened some problems casued by the global financial crisis of 2008-2011. However, the observed spread increase was mainly due to global uncertainty (Pan and Singleton, 2008; Kocis, 2013; Geršl, Lešanovská, 2013). Neither the counterparty credit nor liquidity risk was significantly noticed in these markets during this time. Moreover, most of these markets were exhibiting the surplus liquidity. Nevertheless, the regulators decided to intervene on the interbank markets to counteract potential further consequences of the global crisis. The set of policy measures in the banking sector was wide; however, central bank interventions dominated. For example, policymakers decided to run repo operations and extend the collateral framework. In addition, many of them decreased the reserve requirements and eased the interest rates. A few of them injected some financial instruments aimed at restoring general confidence in the financial market. Given that interbank markets in emerging countries constitute an important channel of monetary policy transmission, the behavior of interbank markets should be of great interest to policymakers.[1]

Because of the different nature of the financial crisis in emerging markets than in the advanced countries and, in many instances, the precautionary nature of policy interventions in the banking sector, we analyze the impact of the financial crisis on interbank markets and test how effective policy announcements were in restoring banking and country stability in

[1] For an overview of major channels of monetary policy transmission, please see a report by the Bank for International Settlements (BIS) (2008).

emerging countries. More specifically, we ask the following questions: how did the interbank markets in emerging countries react to spillover effects from global economies? What was the effect of policy interventions on interbank and country premiums? And, finally, what was the role of international interventions, which were also introduced in many emerging countries? Surprisingly, though the existing literature examining the effect of various policy responses in advanced countries is huge, there are, to the best of our knowledge, no empirical studies investigating these questions in less advanced countries. Therefore, with our study, we contribute to three arrays of literature. First, we look at the interbank market as a potential shock transmission channel in emerging countries and try to explain its behavior between 2007 and 2011. Second and more importantly, we evaluate the effectiveness of policy responses in the banking sector undertaken in these countries during the global crisis of 2007-2011. Moreover, the results of our study can extend the global literature on the effectiveness of various policy measures in restoring the stability of the banking sector. With our study we test how the policy instruments work under different settings - markets without significant counterparty risk exhibiting surplus liquidity. Since the existing literature present inconclusive results on the effectiveness of policy instruments, we might add to the explanation, why some instruments do not work during the crisis.

We answer our questions as follows. First, we evaluate how interbank market spreads have reacted to the global events between 2007 and 2011 in six important countries in Central and Eastern Europe – Poland, Hungary, the Czech Republic and three Baltic states: Latvia, Lithuania and Estonia. We try to explain the reasons for these changes. To this end, we use the spread between a country‘s Interbank Offered Rates (IBOR) and Overnight Index Swaps (OIS), the spread between the Overnight Index Swaps and Government Bonds of the same maturity and the Credit Default Swap (CDS) premium denominated in U.S. dollars. Second, we construct a detailed database of macroeconomic and financial sector policy initiatives announced in all six of these countries between March 15, 2007 and December, 21, 2011. The database covers both domestic and international announcements in the area of monetary policy (interest rate decisions, and quantitative and credit easing), liquidity support (in domestic and foreign currency), and financial sector policy (mostly liability guarantees and deposit insurance). Finally, following the methodology of Ait-Sahalia et al. (2012) – an event study – we assess how successful macroeconomic and financial sector policy initiatives addressed distress in the banking sector. We employ parametric means tests to evaluate whether policy announcements had an economically and statistically significant impact on interbank risk premiums.

In theory, it is difficult to expect any clear impact from policy actions on the recovery of the emerging countries. On the one hand, we might expect negative effects for such actions because in markets with surplus liquidity, central bank actions can break down the transmission mechanism of monetary policy. If there is effective monetary policy implementation, changes in central bank rates are fully and rapidly reflected in changes in interbank rates and then in the wider structure of rates in the banking system. This is especially underpinned by the existence of a liquidity shortage in the interbank market because it forces the market into transactions, which allow the central bank to exploit its position as the monopoly supplier of domestic currency (Ganley, 2004). In turn, when the markets experience surplus liquidity, the transmission mechanism can break down because the banks have discretion as to whether they lend their surplus reserves. The central bank cannot force the banks to lend to it, so to one extent or another, the central bank's ability to transmit its preferred interest rate into the market is weakened. Second, when the spreads are not driven by counterparty risk but rather by other external factors, for example general uncertainty, actions by the central bank may even intensify the risk aversion of banks, as they might signal „bad information". Consequently, Brunetti et al. (2011) document that central bank interventions create greater volatility and reduce the trading volume of the interbank market. Further, this may lead to greater risk aversion and precautionary motives for liquidity hoarding by banks (Angelini et al., 2011; Acharya and Merrouche, 2012; Heider et al., 2015). As a consequence, the interbank spread will increase. On the other hand, some academic evidence demonstrates that policy interventions have a stabilizing effect on the interbank market, and if they had not been run, the spreads would have been much larger (Christensen et al., 2009; Ait-Sahalia et al., 2012). In particular, international interventions might be positively viewed by the market due to greater creditworthiness of international institutions and potential financial support they can afford to (Papi et al., 2015).

Our results show that the interbank markets in emerging markets, based on a sample of CEE countries, have been affected by the global financial crisis, especially following the bankruptcy of Lehman Brothers in Sept., 2008. As a result, spread measures increased. The policy interventions, which should have stabilized the market, induced an even further increase in interbank premiums. We document that interest rate reductions and liquidity support measures in particular increased all interbank distress indicators in our sample countries, namely the IBOR-OIS spread, swap spread and CDS spreads. This seems to be consistent with the view that in markets where liquidity is not an issue, central bank interventions are not effective in stabilizing the interbank market (Brunetti et al., 2011). In turn, such actions might signal “bad news”, leading banks to precautionary liquidity hoarding. Frontloading and underbidding were very popular after that time (Lu, 2012). Among the liquidity instruments, we show that especially repo negatively impacted the interbank market the day after the intervention. This might indicate that in markets where liquidity is not a concern, banks have discretion about how to manage their surplus reserves; thus, central bank actions do not fulfil their goal. A good example is Poland, where the central bank has even stopped its operations due to reduced demand (Lu, 2012). Moreover, we do not find any statistical evidence of financial sector policy instruments having an impact on the spread measures, which might indicate a positive signal. Nevertheless, given that such instruments were not widely used by our sample countries, this effect should be interpreted cautiously. However, some weak evidence suggests that the international financial sector policy might help markets stabilize their situation, as the banking sector positively assessed the involvement of international institutions. As a result, the IBOR-OIS spread seems to decrease. Finally, we notice the spillover effects of policy interventions on spreads in other countries. Moreover, we find that the spillover effects between our sample countries were significant, especially after the announcements of central bank actions. This is because banking sectors in CEE countries are dominated by the same foreign capital, which drive similar behavior between the banks. This is in line with studies of Kocsis (2013) and Kliber and Pluciennik (2014), who document that uncertainty increased the risk correlation between neighborhood countries in CEE region. Our results are robust with regard to five-day and three-day event windows as well as across our interbank distress measures. They also stay the same if we exclude countries more exposed to ECB annoucements.

The paper is organized as follows. The second section presents the contribution of this study to other existing studies. The third section describes the data, whereas the fourth section describes the methodology. The fifth section presents the empirical findings. And finally, the sixth section concludes the paper.

2. Contribution to the existing literature

With our study, we contribute to three arrays of the literature. First, we examine how financial shock has affected the interbank markets in emerging countries and, more specifically, Central and Eastern Europe. Surprisingly, the existing literature on the interbank market as a channel for shock transmission is almost non-existent. Though this fact may be attributed to the poor development of this market in emerging countries, compared to advanced countries as well as its limited role in the transmission of liquidity, the literature on emerging markets has shown that the interbank market is an important channel of monetary policy transmission (see a study of BIS, 2008). Thus, it should be of interest to policymakers to evaluate the effect of the global events of 2007-2011 on the condition of the interbank markets in these countries. However, the existing literature evaluates financial shock transmission between developed and emerging countries mainly through a reduction in trade, limited capital outflow and restricted capital access (Eichengreen and Rose, 1999; Glick and Rose, 1999; Forbes, 2001; Kaminsky and Reinhart, 2003, Caramazza et al., 2000; Fratzscher, 2002; Van Rijckeghem and Weder, 2001; Garcia-Cebro et al., 2011; Fink and Schueler, 2015). The banking sector as a channel of financial shock transmission has received less attraction, though interest has risen recently. Peek and Rosengren (1997) document that Japanese banks reduced their lending in the U.S. as a result of the financial crisis in Japan. Similar results were found by Schnabel (2012) for the Russian crisis. Using data on the Russian crisis of 1998, the authors show that international banks reduced their lending to Peruvian banks after they were affected by the Russian financial shock. The effect was the strongest for Peruvian banks that borrowed internationally. More specifically, De Haas and van Lelyveld (2010), Cull and Martínez Pería (2013) and Allen et al. (2014) document that lending by foreign bank subsidiaries depends on the financial strength of the parent bank. Consequently, the authors show that foreign-owned subsidiaries relying on financing by their parent reduced their lending during a crisis as a result of parent problems. Thus, the authors conclude that internal capital markets might be an important channel of financial shock transmission between countries. The financial crisis of 2007-2011 has also raised the importance of domestic interbank markets as a source of contagion in and between countries. Only a few studies on emerging markets look at domestic interbank markets as a contagion engine for financial shock transmission. Most of them, however, concentrate on the determinants of CDS spreads. For example, Doodley and Hutchison (2009) analyze the effect of U.S. subprime news on changes in CDS spreads in 14 emerging countries. The authors find that U.S. financial and real news transmitted strongly to emerging markets over the whole sample period, as reflected in 5-year CDS spreads on sovereign bonds. They find that a wide set of U.S. news announcements, such as write-downs of financial institutions and news on the U.S. real economy, systemically moved CDS spreads in most emerging markets. However, Lehman Brothers' bankruptcy and swap arrangements had uniformly large effects across all of the emerging markets in the sample. By contrast, major news announcements by the Federal Reserve and U.S. Treasury on plans to stabilize the U.S. financial system had little effect on emerging market CDS spreads. Pan and Singleton (2008) or Longstaff et al. (2011) document that the change in CDS spreads in emerging countries can be mostly attributed to regional and global factors; idiosyncratic country-specific variation represents only a small fraction of the total variation in sovereign credit spreads. Similar results are presented by Kocsis (2013) for Hungarian CDS. In addition, the author documents that global factors increased the correlation through which Hungary and peripheral countries are correlated, whereas regional factors were weak channels of co-movement. Just one study looks at the interbank market and its measures of distress specifically. Geršl and Lešanovská (2013) analyze the interbank market premium, defined as the difference between 3M PRIBOR and 2W repo, in the Czech Republic during the global financial crisis. The authors find that foreign market risk as well as the risk of the domestic bond market played important roles in determining interbank money. Liquidity risk was not important; however, counterparty risk seemed to have an impact first after the period of 2009. With our study we look at the interbank markets in emerging countries during the global crisis of 2007-2011. Though we do not empirically determine the spread determinants, we try to give some insights of functioning and spillover effects coming from the advanced countries into the interbank markets in the emerging countries.

To counteract potential problems from the global economies entering emerging countries, regulators decided to react in the financial sector. With our study, we assess how effective these actions were in stabilizing banking sectors in CEE countries. This is the first study that tries to assess the impact of financial sector initiatives in emerging countries. All existing studies mostly concentrate on advanced countries – especially the USA, Japan, the UK, and the euro area. For example, Taylor and Williams (2009), Michauld and Upper (2008), Frank and Hesse (2009) and Pasquariello et al. (2014) examine the effect of Fed interventions on the interbank market in the USA. Ait-Sahalia et al. (2012) investigate the influence of financial policy initiatives on the interbank markets in four developed regions: the USA, the UK, Japan and the euro area. Similarly, Cecioni et al. (2011) review the impact of non-conventional measures on financial and macroeconomic variables using the U.S. and the euro area. Acharya and Merrouche (2012) present evidence on the effect of interventions by the Central Bank of England on interbank market in that country. Other studies concentrate solely on the euro area. For example, Brunetti et al. (2011) and Ghysles et al. (2013) examine the effect of European Central Bank (ECB) interventions in the so-called "Securities Markets Programme" on stability of the euro government bonds of countries included in the program. In contrast, Hesse and Frank (2009) review the effect of ECB funding liquidity on spread depression. The only study that examines the impact of policy interventions on the financial sector was performed by Doodley and Hutchion (2009). However, the authors mainly evaluate the impact of macro and fiscal reforms on the reduction of CDS spread in 14 selected emerging economies. More specifically, they analyze the following instruments used to counteract the spillover effects from the U.S. market into these markets during the global crisis: increases in reserve assets, reductions in net government debt and the foreign currency exposure of governments, restrictions in commercial bank net foreign exchange borrowing, and strict monitoring of nonfinancial firms' foreign currency.

Third, our study also contributes to the general literature on the effectiveness of various policy interventions during the financial crisis. This literature has grown significantly in the recent years; however, it presents mixed evidence, failing to present unambigous policy conclusions. For example, Taylor and Williams (2009) and Stroebel and Taylor (2012) document that U.S. Federal Reserve interventions were not effective in reducing counterparty risk, defined as the spread between LIBOR and OIS. Similar results were found by Brunetti et al. (2011) analyzing interventions by the European Central Bank. The authors conclude that central bank actions are not successful in mitigating asymmetric information problems and thus restoring confidence in a market. As a result, market liquidity does not improve. Furthermore, the authors also document that central bank interventions may consistently create uncertainty in the interbank market. However, Angelini et al. (2011) document that refinancing operations run by the ECB did not have a large effect on the LIBOR-OIS spread, suggesting that they are not successful in addressing the problem of risk aversion, when it is prevalent in the market. Similarly, Heider et al. (2015) argue that the same applies if banks decide to precautionarily hoard liquidity. Central bank interventions will not solve this problem because banks have discretion regarding participation in the program. Fiordelisi et al. (2014) analyze the effect of conventional and non-conventional monetary policy interventions between July 2007 and June 2012 in the most advanced monetary areas (the euro area, Japan, and the U.S.) on the interbank market, considering the 3-month LIBOR-OIS spread. The authors show that standard measures, such as a drop in interest rates, have been more effective in reducing the spread than unconventional ones, such as monetary easing and liquidity provisions. In turn, other studies document that policy interventions are successful and dampen the spread in interbank markets. For example, McAndrews et al. (2008) and Fleming et al. (2010) document that the Fed's Term Auction Facility (TAF) and Term Securities Lending Facility reduced the LIBOR-OIS spread and repo spreads between the treasury and liquid collateral, respectively. Interestingly, Christensen et al. (2009) using the factor model, show that if not the TAF intervention, the LIBOR spread would have been much higher, concluding that such actions are necessary. Hesse and Frank (2009) and Angelini (2011) argue that the initial rise in the spread was due to funding liquidity and provide evidence that supplementary long-term refinancing operations by the ECB and the TAF by the FED did compress the LIBOR spread. Similarly, Lenza (2010) document that the effect of the easing maneuver on the interbank spread was sizable and demonstrate the expansionary impact on the economy. Cecioni et al. (2011) review the impact of non-conventional measures on financial and macroeconomic variables with reference to both the U.S. and the euro area and find that generally these measures were effective in reducing interest rate spread. Interestingly, Ait-Sahalia et al. (2012) examine the effect of policy interventions (fiscal and monetary policies, liquidity support, financial sector policies and ad hoc bank failures) on interbank credit and liquidity risk premiums in the U.S., the euro area, the UK and Japan between 2007 and 2009. They use LIBOR-OIS and show that policy announcements were usually associated with reductions in the LIBOR-OIS spreads, but there is no policy action that is better than others for containing the crisis. Overall, the existing studies mostly concentrate on how policy initiatives affect liquidity and counterparty risk premiums, addressing interbank liquidity and asymmetric information problems. We add to this literature by testing how policy instruments respond under different settings, as the spread movements in CEE countries between 2007 and 2011 were different in nature (Lu, 2012; Plociennik, 2012).

3. Measuring financial sector distress and policy initiatives

3.1. Measuring financial distress

We study the short-term response of interbank credit and liquidity risk premiums to different types of policy announcements in six Central and Eastern European countries—Poland, Hungary, the Czech Republic, Lithuania, Latvia and Estonia—using the event study methodology.

Following the existing literature, we use a set of interbank risk premium measures. First, we use the spread between a country's transaction-based three-month interbank offered rate (IBOR) and overnight interest swap (OIS). We calculate the spread as the difference between accurate three-month interbank offered rates and the three-month overnight index swaps. The IBOR rates for individual countries are as follows: WIBOR for Poland, WUBOR for Hungary, PRIBOR for the Czech Republic, and EURIBOR for Lithuania, Estonia and Latvia due to their connections to the euro zone.[2] The existing literature uses this type of spread to measure the impact of counterparty and liquidity risks on the interbank market as well as to assess the impact of various events on this market (McAndrews et al., 2008; Schwarz, 2009; Poskitt, 2011; Ait-Sahalia, 2012). Moreover, the ECB, using the LIBOR-OIS spread, assesses the effect of confidence regarding the crisis on European, American and British markets and find the correlation between individual markets. The Soultanaeva and Strömqvist (2009) run similar studies for Sweden. The authors test how the American and euro crises spread to the Swedish markets. We also use the swap spread. Specifically, we define the swap spread as the spread paid by the fixed-rate payer of an interest rate swap (IRS) over the rate of the run treasury with the same maturity as the swap. Consequently, we take the five-year IRS paid on domestic currency and treasury yields with the same maturity. Duffie and Singleton (1997) document that the main determinants affecting the change in the swap spreads are liquidity and counterparty risk. In addition, Sorensen i Bollier (1994), Lekkos and Milaas (2001, 2004), Choudhry (2008), Huang et al. (2008) and Van Deventer (2012) document that interest rate expectations and investors' perception about the riskiness of government bonds might change the swap spread as well. Thus, whereas the IBOR-OIS spread specifically captures the risk of counterparties in the interbank market, the swap spread is a broader measure of riskiness, explicitly capturing the riskiness of a country. Finally, we also consider a composite measure of bank-specific default risk known as the credit default swap (CDS) spread based on USD denominated treasury bonds. The CDSs for CEE countries have been traded since October 9th, 2008.

[2] Until 2011 Latvia had a pegged currency regime toward euro, however in 2011 it adopted the euro currency. Lithuania and Estonia have introduced the currency regime board toward euro (Purfield and Rosenberg, 2010).

We analyze the impact of policy announcements on changes in the studied variables and capture the cumulative impact of policy announcements over a few days. The methodology of event study requires the aggregation of abnormal differences in variables within each event window to construct cumulative abnormal differences (CAD), assuming that no other factors occurred in that time. The data gathered for this research are taken from the Thompson Reuters Eikon system.

3.2. Policy announcements in Central and Eastern countries

We compile data on major policy initiatives announced by country authorities in Central and Eastern Europe, covering Poland, Hungary, the Czech Republic, Lithuania, Latvia and Estonia. The dates of policy announcements were identified based on the central banks' reports and websites. When identification from public source was not possible, we contacted central bank officials directly to get informed about the forms of banking interventions in a given country. We dated policy announcements as of their official announcement. Therefore, our data are daily.

We classify announcements into domestic and international initiatives. The domestic announcements include (i) monetary policy, (ii) liquidity support, and (iv) financial sector policy, whereas international initiatives include (i) liquidity support and (ii) financial sector policies, mainly in the form of access to IMF credit lines.

Monetary policy measures include decisions concerning changes in the interest rate. For our analysis, we consider the date of the announcement.

Liquidity support is the provision of currency liquidity through broadened access to central bank refinancing, an extended collateral framework, more frequent actions, or longer maturities. Domestic liquidity support is further broken into repo actions, changes in reserve requirements and collateral easing. International liquidity support includes foreign currency liquidity through foreign swap agreements between the central banks of CEE countries and the European Central Bank, the Bank of Sweden and the Bank of Denmark.

Financial sector policies include the tools commonly utilized to solve systemic banking crises. For Central and Eastern Europe, this policy intervention mainly includes the introduction of a

Financial Stability Committee and changes in deposit protection schemes. International policy, however, includes access to the flexible credit lines of the IMF granted to some CEE countries during the crisis. The purpose of the credit lines is to protect the confidence in a country before it goes to external markets in search of financing.

In total, the database includes 112 announcements (Table 1).

[Table 1]

Monetary policy announcements account for the largest share of intervention announcements in CEE countries (76), followed by liquidity support interventions (17), and financial policy announcements (3). This is an opposite situation to that in advanced countries, where financial sector policy initiatives accounted for the largest policy actions (Ait-Sahalia et al., 2012). The most announcements were made by Poland (31), Hungary (33), and then the Czech Republic (17).

Liquidity injections were accompanied by a decrease in interest rates, which occurred throughout the global crisis period in all CEE countries. Our analysis for monetary instruments excludes Estonia, which was in a currency board arrangement.[3] In most of the analyzed countries, the largest changes in interest rates occurred in 2009 as a result of global distress. Two countries, Lithuania and Poland, decided to react by decreasing their interest rates more dramatically as a reaction to the collapse of Lehman Brothers. Hungary changed its interest rates less drastically but more frequently. One third of all 76 monetary policy announcements occurred in this country.

In terms of other policy announcements, we observe that the most liquidity announcements occurred in Poland and Latvia. The change in reserve requirements was the most common central bank intervention in these countries. This is not surprising because such an action should immediately improve the liquidity position of the banking sector. Other common instruments were repo operations and collateral easing. Some countries decided to introduce all of these policies, such as Poland, whereas other countries concentrated on one of them, such as Latvia. In Estonia, the central bank did not decide to intervene locally at all. It entered into an FX swap agreement with the Bank of Sweden; however, the instrument has not been

[3] The Estonian monetary system was based on a fixed exchange rate regime against the euro with currency board arrangement. Due to the currency board arrangement, the Bank of Estonia did not implement independent monetary policy. The money supply in Estonia was endogenously determined by money demand. The central bank did not implement open market operations and did not influence the money supply by selling bonds or by setting interest rates.

used. Other countries also went into FX swap agreements with foreign institutions to protect the liquidity of the interbank market. This was especially needed in the FX market, whose liquidity deteriorated as a result of Lehman Brothers' collapse and the lack of confidence in global markets. However, the need for foreign currency was a result of the banking sector's loan portfolio, which in most of the CEE countries, was held to approximately 60-70 percent foreign currency. In many cases, the FX swap agreements were precautionary in nature. Poland and Hungary entered into an ECB swap line, and Latvia did so with the central banks of Sweden and Denmark. Though these actions were precautionary in nature, it should be expected that their announcements had an important effect on the behavior of banking institutions. In the later stage of the crisis, mainly as a result of the euro debt crisis, some countries decided to cooperate with the IMF to protect their creditworthiness. Poland, the Czech Republic, Hungary and Latvia decided to apply for a flexible credit line at the IMF. Only Poland injected any domestic financial policy instruments.

Table 2 summarizes the policy instruments used during the subprime crisis in five CEE countries.

[Table2]

4. Event study methodology

We evaluate the real-time response of interbank credit and liquidity risk premiums to policy announcements using the event study methodology. The methodology of event study is well established in the finance literature (see, for example, Campbell et al., 1997; Kothari and Warner, 2007; Ait-Sahalia et al., 2012). It is generally well suited to assessing the short-run response to policy announcements. It is quite simple and allows us to work with the limited number of announcements we have. In addition, it avoids the specification issues underlying the spread model.

An event study needs to be designed carefully to address several issues. To create multiple sets of similar events, we classify announcements into the few policy event types discussed in the previous section. Then, we pool them within each sample of event type across countries. More specifically, we perform country-specific analyses, using the various spread measures in respective currencies to gauge the extent to which the spread changes as a result of specific country announcements. The country results are then pooled across the same policy measures.

It is also important to ensure that the results in a specific pool are not influenced by other events. Moreover, similar to Ait-Sahalia et al. (2012), to minimize the endogeneity problem and to ensure that the effects of multiple announcements do not contaminate the results, we focus on major non-overlapping policy announcements. Therefore, when doing the analysis on news for each pool, we exclude announcements that fall within five days from each other except for announcements linked to different policy measures. We also exclude concurrent events on the same day for which identifying the main event was difficult. Such a situation, however, happens for only two observations.

In our study, we use three different pooling conceptions. The first one is focused on combining both domestic with international interventions into one pool. The second considers domestic and international support announcements, independently. Finally, in the third pooling, we test the effectiveness of domestic liquidity instruments on the interbank market in CEE countries, distinguishing the individual effects between such liquidity instruments as follows: repo actions, changes in reserve requirements and collateral easing.

Limiting the size of the event window helps to avoid any noise coming from other news or events. Therefore, similar to Ait-Sahalia et al. (2012), we use an event window of five days: one day before and three days after the announcement date. We also present the results using the symmetric three-day window, one prior and one after the event date. However, most of the results are unchanged.

Consistent with McQueen and Roley (1993), we expect the market response to announcements to be state contingent, i.e., to depend not only on the surprise content of the announcements but also on the state of the economy and financial markets in which investors interpret them. Though the event study does not fully control for the economic conditions that may affect the market response to news, our analysis mostly relies on interventions between and 2010 (just a few were happening in 2007 and 2011), capturing the period after Lehman Brothers' collapse. Therefore, we argue that we capture the same market sentiment that existed after this period.

In the event study methodology, statistical tests are based on abnormal differences, which are computed as the difference between the actual daily spread value on each day of the event window and the expected spread measured as the average daily spread over the previous 20 working days of the estimation window. In this way we obtain abnormal differences, which we test using a parametric approach. Parametric tests attribute an equal chance to achieving both positive and negative deviations from our expectations. However, a small number of observations may weaken the power of statistical tests, suggesting the need to consider both the economic and statistical significance of results.

4.1. The evolution of spreads in CEE countries

The deteriorating quality of bank asset and the realized losses from toxic investments have severely impacted the liquidity of the interbank markets in most advanced countries. Interbank spreads have significantly reacted to this situation and increased to a level previously not seen. The existing literature assigns this fact mainly to the counterparty risk and liquidity risk (McAndrews et al., 2008; Gorton, 2009, Heider et al., 2015, Michaud and Upper, 2008, Schwarz, 2009; Taylor and Williams, 2009). As bank accounts deteriorate, increased risk leads to adverse selection problems that may provoke interbank market failure. Consequently, banks unsure about the depth of the problems on other banks' balance sheets are simply unwilling to lend to each other without substantial accommodations for the risk. However, we expect a different nature for interbank markets in emerging countries, where neither counterparty risk nor liquidity risk exists. Nevertheless, the interbank markets of these countries have also severely reacted to global turbulence. The question is how and why interbank markets have been impacted by the situation of global banks.

Figures 1-3 present the evolution of three interbank financial distress measures in six CEE countries between 2007 and 2014: the swap spread, the IBOR-OIS spread and the credit default swap spread. However, we are mostly interested in the spread change between 2008 and 2011.

[Figures 1-3]

The data show that all three spread measures reacted to the situation in the global market, especially after the bankruptcy of Lehman Brothers and during the euro debt crisis. The spreads before Sep. 2008 were quite stable, though their level was slightly higher compared to what we had observed recently. This might be an indication that the global crisis was barely evident in CEE countries before that time. Consequently, until that time, we did not observe any policy initiatives in these countries. This situation changed in the second half of 2008, when CDS and LIBOR-OIS spreads significantly increased, as a consequence of the troubles of financial institutions such as Bearn Stearns and Northern Rock, and finally the bankruptcy of Lehman Brothers in September 2008. Figure 1 shows interesting results regarding the swap spread. Their values turned negative. Huang et al. (2008) and Van Deventer (2012) document that this movement is associated with a change in the treatment of treasury bonds, which stopped being considered a risk-free investment by investors. The risk of spillover effects coming from global economies into emerging countries, mostly related to macroeconomic factors, such as the outflow of foreign investors, lower international demand or restricted capital access to international markets, have hampered the confidence in CEE countries, increasing the yields on government bonds. In addition, the general situation in the European Union has also impacted the treatment of government bonds in CEE countries by investors because investors generally tend to consider the region as a whole rather than countries individually (Arghyrou and Kontonikas, 2012; Attinasi et al., 2009). Thus, the credit risk of some European Union countries has affected the treatment of other countries, including CEE economies. Moreover, expectations concerning the monetary policy have also played an important role in determining the swap spread in emerging countries. In fact, Pluciennik (2014) finds that the yield curve has played the most important role in determining the swap spread in Poland. In turn, he does not observe a relation between the change of swap spread in Poland and the euro or US swap spread.

Concerning the credit default and IBOR-OIS spreads, we find that between mid-2008 and mid-2009, the largest increase occurred in the Baltic states (Figure 2 and Figure 3). This period reflects the beginning of the debt crisis in Europe. All three Baltic countries were closely connected with the euro currency, and investors were worried about the risk of the Baltic states during the deteriorating European situation. The data on credit default swap spreads seem to support our argumentation. CDS spreads reflect the fear of investors about a country's risk, and we notice the highest premiums for the Baltic states. In contrast, Poland and the Czech Republic experienced the lowest movements in CDS spreads. In addition, the premium increase in all CEE countries was related to the deterioration of the credit portfolio of parent banks whose subsidiaries were located in these countries. Investors were worried about the financial shock transmission from parent banks to their subsidiaries (Allen et al., 2014).

The IBOR-OIS (Figure 1) followed the same pattern as CDS spreads, resulting in a significant increase after Lehman Brothers' bankruptcy and during the euro zone crisis. Nevertheless, this spread should rather reflect the credit and liquidity risk in the interbank market

(McAndrews et al., 2008; Gorton, 2009; Michaud and Upper, 2008). Because these two types of risks were not a concern for the banking sectors in CEE countries, the increase came as a surprise. Interestingly, Kliber and Pluciennik (2011) document that the spread development in Poland was independent from the euro and US spread development. It followed the same trend, but with some delay. We argue that global uncertainty probably made domestic banks more risk averse, which resulted in the liquidity hoarding behavior of these banks. The unwillingness of banks to lend money has increased the cost in interbank markets, especially over a longer period. Risk aversion and liquidity hoarding have also been identified in the financial crisis literature as important factors for the breakdown of interbank markets in advanced countries (Angelini et al., 2011; Acharya and Merrouche, 2012; Heider et al., 2015).

5. Impact of policy announcements on interbank credit and liquidity risk

Following the decreased confidence in the interbank market in CEE countries as well as global financial turbulences, most policy initiatives took place between 2008 and 2011. During this time, the central banks decided immediately to decrease required reserves and induce repo operations to counteract the potential negative consequences of the subprime crisis. Many of them also eased collateral requirements. All of these actions were accompanied by a decrease in interest rates in all CEE countries. At the same time, CEE central banks agreed to enter into FX swaps with the ECB and other central banks.

To gauge the response of various spread measures to announcements of policy interventions in a more formal way, we first plot changes in the IBOR-OIS spreads and then test the statistical significance of differences in the behavior of the IBOR-OIS spreads prior to and after the announcements. We start with a graphical analysis and then we come to the statistical tests on a pooled sample of announcements and other spread measures. We then proceed to the analysis of spillovers from policy announcements.

5.1. Graphical analysis

Figures 4-7 present the cumulative effect of four groups of interventions on changes in the IBOR-OIS spread in a five-day window on a pool of six CEE countries. The interventions are as follows: (4) domestic financial sector policy, (5) domestic liquidity support, (6) international financial sector policy, and (7) international liquidity support. The vertical axis presents the days around interventions, with zero being the day of the intervention.

[Figures 4-7]

In general, the results document that announcements of policy interventions resulted in an increase in interbank spread during the crisis of 2007-2011 in all CEE countries. The instruments in the range of international financial sector policy are the only exception. Furthermore, the data suggest that the dynamics of spread changes as a result of intervention announcements are more favorable to international than domestic announcements. The spreads after international announcements decreased more drastically.

Moreover, we also find that among the domestic announcements, the financial sector instruments seem to be less effective than liquidity instruments, as the increase in the spread was much higher after the former intervention. On average, we find that the spread after the implementation of financial sector policy instruments increased by 0.93 bp over three days, whereas the spread after liquidity announcements increased by 0.39 bp over the same period. Moreover, the increase in the days following the announcement was much lower than that following financial instruments' implementation. The increase in spread after liquidity instrument announcements is in line with studies from advanced countries. Ait-Sahalia et al. (2012) document that the spreads in advanced countries increased after liquidity announcements during the global crisis phase but not during the subprime period. Taking into consideration that most of the interventions in CEE countries (or at least the most powerful ones) occurred during the global phase of the crisis, this result can be viewed as consistent with other studies.

Interestingly, we see that the spread after international liquidity announcements increased; however, the dynamics were lower compared to the period before them. On average, we find that the spread increased by 0.68 bp as a result of international liquidity announcement over a three-day period, which is more than that after domestic interventions.

In contrast to other instruments, international financial sector instruments lead to a decrease in spreads after the announcement. This decrease was 0.03 bp. The results seem to suggest that the announcement of flexible credit lines by the IMF seemed to stabilize the interbank market, which became very nervous during the euro debt crisis, especially between 2009 and 2011.

5.2. Statistical test

The above analysis seems to raise doubts about the effectiveness of most of the policy initiatives aimed at stabilizing the situation in CEE countries, at least in the short run. In this sub-section, we test the statistical significance of the effect of policy announcements on spread measures. Moreover, we analyze the differences between the effects depending on the type of policy instruments. Table 3 presents the effect of policy announcements dividing them into liquidity and financial sector measures. Specifications (1), (2) and (3) present the estmation results for five-day event window, whereas specifications (4), (5) and (6) for three-event window, for both domestic and international types of actions combined together as well as seperately for domestic and international measures, respectively. Moreover, among the domestic interventions, we also distinguish between various liquidity instruments in a given country.

[Table 3]

The results for specification (1) and (3) seem to support our previous observation that policy actions were not successful in stabilizing the interbank market in the sample countries. We document that in general, interbank spreads consistently increase after the policy interventions. The result seems to document that the interventions were associated with "bad news" and thus might lead to greater risk aversion and liquidity hoarding behavior, increasing the spreads (Angelini et al, 2011; Heider et al., 2015). Although greater returns may have been achievable in the market, banks preferred to pay a risk premium in return for security, mainly using central bank deposit facilities. For example, in October 2008, the National Bank of Poland decided to stop its open market operations because the supply of bills was higher than the demand for them (so-called underbidding). In addition, the frontloanding of banks' fulfillment of reserves requirements played an important role in the crises in these countries (Lu, 2012). However, the results also show that among the measures undertaken by central banks, only liquidity support seems to have a statistically significant effect on the spreads. The result is also robust among all of the spread measures. This is exactly what the recent studies on the global financial crisis have documented. Central bank interventions are effective only when the markets lack liquidity. Given that liquidity was not a concern in these markets, the result are consistent with the view that liquidity instruments do not solve uncertainty problems in the banking markets (Taylor and Williams, 2009; Stroebel and Taylor, 2012; Brunetti et al., 2011; Heider et al, 2015). Risk aversion dominates bank behavior, and as a result banks prefer to accumulate liquidity. Finally, we also observe an increase in premiums in credit default swaps, which might indicate that banking sectors in CEE countries did not experience liquidity problems but were confronted with global uncertainty. As a result, the policy initiatives did not successfully impact the interbank market.

5.2.1. Domestic announcements

Regarding specification (2) and (4), Table 3 presents the effect of policy announcements on the spreads also controlling for the type of instruments used. Here, we distinguish the monetary policy instruments and liquidity instruments, such as changes in reserve requirements, repo operations and collateral easing. Finally, specification (3) presents the effect of international policy announcements.

The results for specification (2) and (4) indicate that monetary policy and liquidity support have a statistically important impact on the analyzed spreads, though the latter has an effect at a ten percent significance level. The largest change can be noticed for liquidity support. These findings seem to support our previous results. In markets where liquidity is not a problem, popular central bank interventions might lead to greater risk aversion in a market. As a result, the interbank spread will increase. Interestingly, Lu (2012) document that neither changes in the EURIBOR-OIS spread nor the interbank credit risk affected the domestic spread in Poland. Therefore, we can conclude that the spread change reflects the precautionary behavior of banks, rather than risk related to the counterparties.

More specifically, we document that the change in the interest rate has also negatively impacted two other measures: the swap spread and the premium on credit default swaps. We do not find any effect of interest easing on IBOR-OIS spread. The change in the spread might market expectations on interest easing. This is consistent with Pluciennik (2014), who document that the yield curve was the main determinant of the swap spread in Poland. Moreover, the increase in spreads might suggest that interest rate easing has been perceived as “bad news” resulting from interventions.

As far as the impact of individual liquidity instruments on spread measures are considered (specification 2.1.), we find that none of the policies have a statistically significant effect on spread changes one dat after the annoucement (specification 5.1). However, we do find a statistically significant effect for repo operations on the swap spread over a one-day window, causing an increase in the spread. In addition, Lu (2012) documents that repo operations in

Poland are associated with spread higher volatility, particularly when the spread between POLONIA and the reference rate is negative. These findings might suggest that the market reacted nervously to the interventions. This would be consistent with the study of Brunetti et al. (2011), who find the same impact of other policy interventions on the spread volatility and, controlling for other factors, attribute this fact to the treatment of intervention as “bad news”. Nevertheless, our impact washes out after the one-day period.

Finally, we also notice that financial sector policy is statistically insignificant on all spread measures. The reason might be that the scale of financial policy instruments was very limited. This fact is surprising given that the crisis in CEE countries was one of confidence rather than liquidity. Moreover, one should also remember that financial policy is aimed at long-term country recovery; we analyze our impact over a short time period, which might be insufficient to assess the effectiveness of such instruments.

5.2.2. International announcements

One of the most important risks to which banks in CEE countries were exposed was related to the lack of access to foreign currency, when the interbank markets in developed countries were frozen. Therefore, many central banks in CEE countries entered into swap agreements with foreign central banks. As mentioned, the National Bank of Poland (NBP) and Hungarian National Bank (HNB) entered into the swap agreement with the ECB, Latvia arranged a swap contract with the Bank of Sweden and Bank of Denmark, and Estonia entered into a similar agreement with the Bank of Sweden (though the swap has not been used). Given that FX loans in CEE countries constituted an important part of the bank portfolios, access to foreign currency through central bank interventions should have been assessed positively.

Moreover, when the debt crisis in the eurozone came in mid-2009, risk in the banking sector of CEE countries increased, as a result of possible financial shock transmission from parent banks to their subsidiaries in the CEE countries. Banks, due to a fear of counterparty risk, might have hoarded liquidity again. This is also observed in the spread evolution in Figures 1-3, where all spreads increased. The flexible line given by the IMF could have stabilized the situation in the banking sector because it might have indicated the capacity of a country to intervene when necessary. Therefore, international financial instruments should positively affect the interbank situation.

Surprisingly, the results on specifications (3) and (6) document a statistically insignificant effect of international instruments on three spread measures in CEE countries, though it should be noted that financial policy negatively impacts the IBOR-OIS when a three-day period is considered and both policy measures negatively affect the spread one day after the announcement (for the first time, though the effects are statistically insignificant). This might suggest that banking sector positively assessed the involvement of international institutions in restoring confidence in it.

5.3. Spillover effects

In the CEE region, owing to a high degree of integration, it is likely that policy initiatives in one country will have a bearing on market conditions in other countries. Table 4 reports the main findings of the statistical tests of domestic and foreign policy announcements on the three spread measures for five CEE countries, whereas Table 5 present the results for three countries, excluding the Baltic states due to their connection to the eurozone. More specifically, we test how the policy announcements in one country have impacted the spread measures in other countries. The estimation was performed for a five-day event window.

[Table 4]
[Table 5]

The findings document that the spillover effects were feasible. The sign of the effects is consistent with our previous findings, showing that the policy announcements have widened the spreads on the interbank market. More importantly, we also find that policy initiatives in one country have led to the widening of interbank spreads in other CEE countries. The results suggest that central bank actions have spillover effects across countries in Central and Eastern Europe. Banking sectors in CEE countries have been dominated by the same foreign capital; thus, banks react similarly to the announcements in different countries. In addition, the region is exposed to the same type of risk factors. This result is consistent with the findings of Ait-Sahalia et al. (2012), who examine the spillover effects between various advanced countries. The authors support that country-specific announcements have statistically significant effects beyond national borders and that these effects intensified as the crisis deepened. For example, the authors document that U.S. and U.K. announcements about the use of unconventional monetary policy were accompanied by reductions in interbank risk and credit premiums for Japan and the euro area. By contrast, announcements of forex swaps by the euro area's and the U.K. authorities had significant positive spillovers in the U.S. (and global) interbank markets. Breşfelean et al. (2011) analyze the effect of changes in the overnight interbank offered rate in the eurozone (EONIA) on interbank rates in CEE countries and find a spillover effects, though different in nature. The lowest magnitude of change was found for Poland.

Interestingly, however, we generally do not find spillover effects on interbank markets across different CEE countries after financial sector policy announcements. The impact of policy announcements in one country on the swap spread and IBOR-OIS spread in other countries is in most cases zero. In turn, we document that financial sector policies have impacted the country measures, such as credit default swaps. This is not surprising because the role of financial sector initiatives is to affect the confidence of investors in a given country. This confirms that investors treat CEE countries as one region and not individually. Surprising is the effect of IMF announcements, which seemed to increase premiums. These effects are even more visible in Table 5, which excludes the Baltic states.

6. Conclusions

With our study, we examine how the interbank markets in emerging countries have reacted to global financial turbulences and test how effective policy instruments have been in stabilizing the banking sector. As a sample, we used six Central and Eastern European countries, considering policy interventions between 2008 and 2011. To analyze interbank markets in these countries, we used three financial distress measures: the IBOR-OIS spread, swap spread and CDS spread. To evaluate the impact of policy announcements on these measures, we applied the event study methodology with three- and one-day periods.

Our results show that surprisingly, the interbank markets in emerging countries were affected by the global financial crisis. The spread increase came as a surprise because banking sectors in these countries neither directly invested in toxic asset nor had exposure to the banks with a toxic portfolio. Given that the interbank spread measures mostly reflect liquidity and counterparty credit risks, the resulting significant increase was not expected. We argue that it was a result of global risk and general uncertainty. Consequently, the results suggest that the interbank market may also be an important channel for financial shock transmission in emerging markets.

More importantly, our results document that policy interventions, mostly dominated by central bank actions, did not appear to be very successful in emerging countries. To counteract the spillover effects from global economies, emerging countries have implemented a wide set of policy measures, including monetary policy instruments, repo, decreasing the required reserve ratio, extending the collateral framework, and international interventions. Most of them, however, are aimed at equipping the interbank markets in these countries with liquidity, which has not been a main concern in these countries. Most of these markets have a liquidity surplus. Consequently, we find that liquidity measures in particular have negatively affected interbank spreads, causing their increase. We argue that central bank interventions do not solve confidence problems and in turn may induce negative behavior among banks. As a result, banks become more risk averse and tend to hoard liquidity. However, our results, though very weak, demonstrate that international financial instruments may bring some stabilization for the interbank markets in emerging countries.

Our study provides important conclusions for policy makers. First, the regulators of emerging countries should also watch the situation on the interbank market. Even if their market is underdeveloped, compared to the markets in advanced countries, its behavior seems to be driven by the global situation. Second, when liquidity is not a concern, financial sector policy instruments aimed at restoring confidence in a market seem to be more effective, with international announcements having an important impact. Finally, because of the limited role of interbank markets in liquidity transmission, standard central bank instruments have limited impacts. Thus, regulators in these countries should develop other policy instruments in interbank markets to react to the global crisis compared to advanced countries.

Table 1: Classification of policy measures

Type	Measures	Countries
Domestic interventions		
Monetary policy	Interest rate cuts	Poland, Lithuania, Czech Republic, Hungary, Latvia
Liquidity support		
Domestic currency liquidity support	Repo operation	Poland (2008), Czech Republic (2008), Latvia (2007)
	Change in Reserve Requirements	Poland (2008, 2009), Latvia (2007, 2008), Lithuania (2008), Hungary (2008, 2010)
	Extension of collateral framework	Poland, Hungary (2009), Lithuania (2008)
Financial sector policy		
Institutional reforms	Creation of Financial Stability Committee	Poland (2007)
Liquidity guarantees	Change in deposit protection scheme	Poland (2008, 2010)
International Announcements		
Liquidity support		
Foreign currency liquidity support	FX Swaps	Poland (2008), Latvia (2008), Estonia (2009), Hungary (2008)
Financial sector policy		
Recapitalization	IMF credit lines	Poland (2009, 2010, 2011), Latvia (2010, 2011), Hungary (2009, 2010)

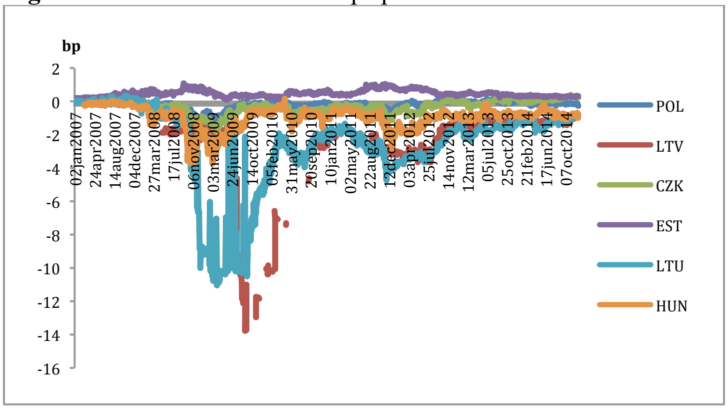
Source: Central banks' resources

Table 2: Number of front-page policy announcements 2007-2011

	Poland	Hungary	Czech Republic	Latvia	Estonia	Lithuania	Total
Domestic policies							
Monetary policy							
Interest cuts	18	26	13	5	-	14	**76**
Liquidity support							**17**
Repo	1		3				4
Reserve Requirements	2	1		4		1	8
Collateral	2	1		1		1	5
Financial Sector Policy	3						**3**
International policies							
Liquidity Support	1	1		1	1		**4**
IMF Credit Lines	4	4	1	3			**12**

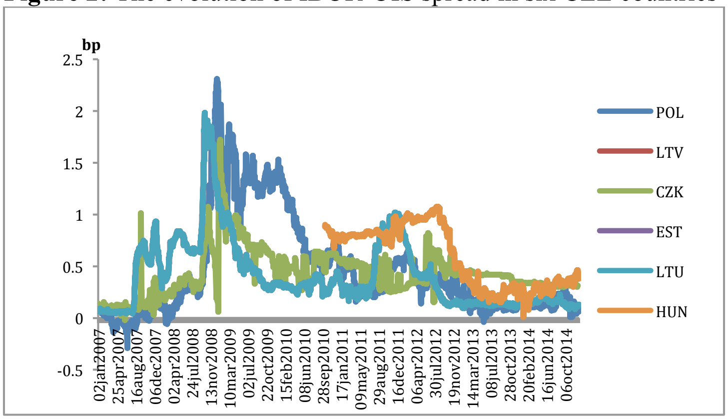
Source: Central banks' resources

Figure 1: The evaluation of swap spread in six CEE countries



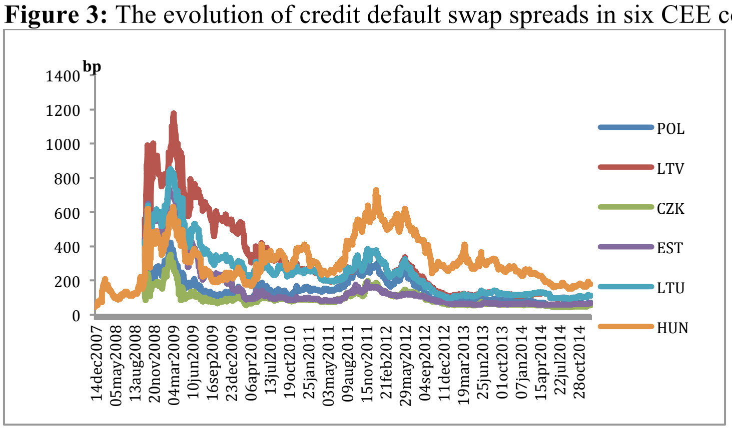
Source: Reuters

Figure 2: The evolution of IBOR-OIS spread in six CEE countries



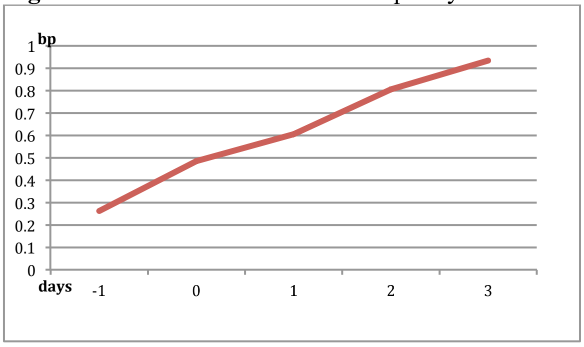
Source: Reuters

Figure 3: The evolution of credit default swap spreads in six CEE countries



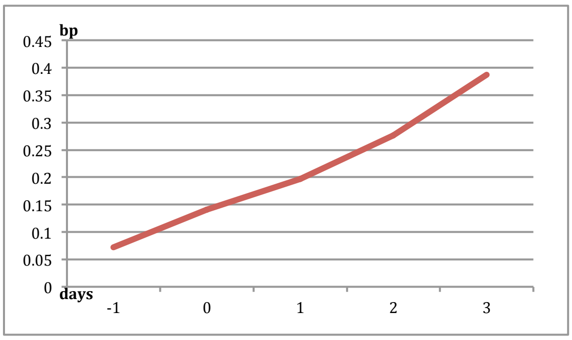
Source: Reuters

Figure 4: Domestic financial sector policy



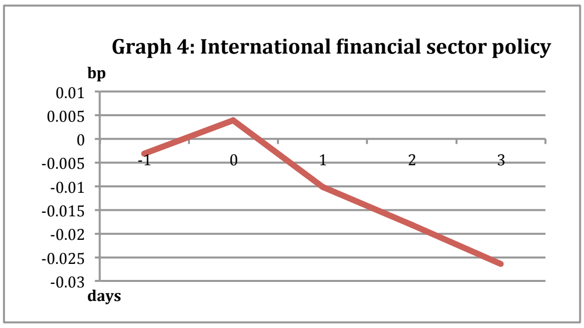
Source: Own calculations

Figure 5: Domestic liquidity support



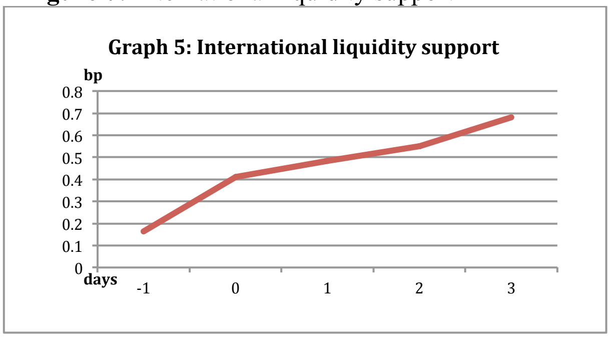
Source: Own calculations

Figure 6: International financial sector policy



Source: Own calculations

Figure 7: International liquidity support



Source: Own calculations

Table 3: Statistical test for alternative event windows

	coef.	t-test	Stat. sign.	obs.	coef.	t-test	stat. sign.	obs	coef	t-test	stat. sign.	obs.
Five-day event window (1-day before and 3-days after announcement)												
	Swap spread				**IBOR-OIS**				**Credit default swap spread**			
Policy announcements (1)												
Fiancial sector policy	-0.232	-0.70		13	0.236	0.93		11	57.87	1.74		14
Liqudity support	-0.769	-2.47	**	16	0.442	1.9	*	16	148.79	2.81	**	14
Domestic Annoucements (2)												
Monetary policy	-0.662	-3.06	***	72	0.17	1.32		54	77.91	2.12	**	52
Financial sector policy	-0.487	-1.25		3	0.934	1.26		3	90.23	1.72		2
Liquidity support	-0.803	-2.01	*	12	0.397	1.99	*	13	154.83	2.19	*	10
Liquidity Support (2.1)												
Reserve Requir.	-0.802	-0.96		5	0.526	1.52		6	177.97	1.55		4
Repo	-0.629	-1.93		4	0.364	0.93		4	3.370	0.12		4
Collateral	-1.038	-1.10		3	0.139	0.83		3	411.46	2.49		2
International Policies (3)												
International Financial sector policy	-0.155	-0.37		10	-0.03	-0.15		8	52.48	1.37		12
International liquidity	-0.667	-1.57		4	0.68	0.64		3	133.69	1.86		4
Three-day event window (1-day before and 1-day after announcement)												
Policy announcements (4)												
Fiancial sector policy	-0.152	-0.74		13	0.157	1.02		11	34.10	1.74		14
Liqudity support	-0.620	-2.50	**	16	0.250	1.62		16	95.94	2.94	**	14
Domestic Annoucements (5)												
Monetary policy	-0.398	-3.19	***	72	0.11	1.38		54	50.91	2.27	**	52
Financial sector policy	-0.260	-1.05		3	0.604	1.35		3	44.17	2.02		2
Liquidity support	-0.671	-2.08	*	12	0.197	1.77		17	94.20	2.18	*	10
Liquidity Support (5.1.)												
Reserve Requir.	-0.897	-1.24		5	0.210	1.15		6	91.63	1.57		4
Repo	-0.367	-3.09	**	4	0.251	0.96		4	-3.75	-0.26		4
Collateral	-0.700	-1.10		3	0.100	0.95		3	295.25	5.62		2
International Policies (6)												
Financial sector policy	-0.119	-0.46		10	-0.010	-0.1		8	32.42	1.42		12
International Liquidity	-0.467	-1.68		4	-0.483	0.62		3	100.30	2.15		4

Table 4: Spillover effects in five Central and Eastern European Countries
The signs of (+) or (-) mean that the spillovers were effective at least at ten precent significance level and contributed to the increase or decrease of the spread, respectively, whereas "0" means that spillovers were not feasible. The spillover effect has been defined as an effect of policy announcement in one country on the other countries in CEE region, based on the pooled country analysis.

	Swap spread	IBOR-OIS	Credit default swap spread
	Policy announcements (1)		
Fiancial sector policy	0	(+)	(+)
Liqudity support	(-)	(+)	(+)
	Domestic annoucements (2)		
Financial sector policy	0	0	(+)
Liquidity support	(-)	(+)	(+)
	Liquidity support (2.1)		
Reserve Requir.	(-)	(+)	0
Repo	(-)	(+)	(+)
Collateral	(-)	(+)	(+)
	International policies (4)		
International financial sector policy	0	(+)	0
International liquidity	(+)	0	(+)

Table 5: Spillover effects for three CEE countries
The signs of (+) or (-) mean that the spillovers were effective at least at ten precent significance level and contributed to the increase or decrease of the spread, respectively, whereas "0" means that spillovers were not feasible. The spillover effect has been defined as an effect of policy announcement in one country on the other countries in CEE region, based on the pooled country analysis.

	Swap spread	IBOR-OIS	Credit default swap spread
	Policy annoucements (1)		
Fiancial sector policy	0	(+)	(+)
Liqudity support	(-)	(+)	(+)
	Domestic Annoucements (2)		
Financial sector policy	0	0	(+)
Liquidity support	(-)	(+)	(+)
	Liquidity Support (2.1)		
Reserve Requir.	(-)	(+)	0
Repo	(-)	(+)	(+)
Collateral	(-)	(+)	(+)
	International Policies (4)		
International financial sector policy	0	(+)	(+)
International liquidity	(-)	0	(+)

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